

Task 1: Understanding Cyber Security Basics & Attack Surface

Cyber security is about protecting our digital information and systems from being misused, stolen, or damaged. Today, almost everything we do depends on technology, like online banking, social media, emails, and mobile apps. Because of this, cyber security has become very important.

1. Meaning of Cyber Security and CIA Triad

Cyber security mainly focuses on three important principles called the CIA triad: confidentiality, integrity, and availability.

Confidentiality means that information should be kept private and only shared with authorized people. For example, when we use a banking app, our password and account details should not be visible to anyone else. Similarly, social media chats are private and should not be read by strangers.

Integrity means that data should remain correct and unchanged unless the right person updates it. For example, the amount of money in a bank account should not change on its own. If someone edits marks in a college system without permission, it breaks data integrity.

Availability means that systems and services should be accessible whenever users need them. For example, if a banking app stops working during an emergency, it becomes a serious problem. Cyber attacks like server overloads can affect availability.

2. Types of Attackers

There are different kinds of people who attack systems, and each has a different reason. Script kiddies are beginners who use already available hacking tools. They usually don't understand how things work deeply and hack mostly for fun or to show off.

Insiders are people who already work inside an organization, such as employees. Since they already have access, they can misuse data, which makes them very dangerous.

Hactivists attack systems to support political or social causes. They may deface websites or leak information to spread a message.

Nation-state attackers are hackers supported by governments. They are highly skilled and target important systems like defense, power grids, or financial institutions.

3. Common Attack Surfaces

An attack surface is any place where an attacker can try to enter a system.

Common attack surfaces include:

Web applications, such as login pages and online forms

Mobile applications, like Android and iOS apps

APIs, which allow apps to communicate with servers

Networks, including Wi-Fi and internet connections

Cloud infrastructure, where data and apps are stored online

More features and connections usually mean a bigger attack surface.

4. OWASP Top 10 Vulnerabilities

The OWASP Top 10 is a list of the most common and dangerous security problems found in web applications.

Some of these include weak login systems, poor access control, SQL injection, exposed sensitive data, and incorrect system settings. These issues are dangerous because they allow attackers to steal data, change information, or take control of systems easily.

5. Mapping Daily-Used Apps to Attack Surfaces

Apps we use daily also have attack surfaces.

For email apps, the attack surface includes the web app, mobile app, server, and network.

For WhatsApp, the mobile app, APIs, cloud servers, and encryption system are all part of the attack surface.

For banking apps, the attack surface includes the mobile app, authentication system, backend server, database, and internet connection.

6. Data Flow: User → Application → Server → Database

In most applications, data flows in a simple way. The user enters information in the app. The app sends this data to a server. The server processes the request and interacts with the database to store or fetch data. Then the response is sent back to the user.

7. Points Where Attacks Can Happen

Attacks can happen at any stage of this flow. Users can be tricked through phishing attacks.

Applications can have bugs or weak validation. Data can be intercepted while traveling through the network. Servers may be misconfigured. Databases can be attacked to steal or delete data.

8. Overall Summary

In simple words, cyber security is about protecting systems and data at every level.

Understanding the CIA triad helps us know what needs protection. Knowing different attackers and attack surfaces shows us where risks come from. OWASP Top 10 helps identify common mistakes in applications. By studying how data flows and where attacks can happen, we can build safer systems and use technology more securely in our daily lives.

Interview Questions

1. *What is the CIA triad?*

The CIA triad is the basic foundation of cyber security. It stands for Confidentiality, Integrity, and Availability.

Confidentiality means data should be private and accessed only by authorized users, like bank passwords or personal messages.

Integrity means data should remain correct and not be changed without permission, such as account balances or exam records.

Availability means systems should be accessible when users need them, for example online banking or email services should not go down.

2. What is an attack surface?

An attack surface is all the possible places where an attacker can try to enter or attack a system.

This includes web applications, mobile apps, APIs, networks, servers, and cloud services.

The larger the attack surface, the more chances an attacker has to find a weakness.

3. Difference between vulnerability, threat, and risk?

A vulnerability is a weakness in a system, like weak passwords or outdated software.

A threat is anything that can take advantage of that weakness, such as a hacker or malware.

Risk is the possibility of damage when a threat uses a vulnerability.

In simple words, vulnerability is the problem, threat is who can use it, and risk is the damage that can happen.

4. What are common cyber attackers?

Common cyber attackers include:

Script kiddies, who use ready-made tools and have limited knowledge

Insiders, who misuse access from inside an organization

Hacktivists, who attack for social or political reasons

Nation-state attackers, who are government-backed and very skilled

Each attacker has different goals and skill levels.

5. Why is OWASP Top 10 important?

OWASP Top 10 is important because it lists the most common and serious security issues found in web applications.

It helps developers and security teams understand what mistakes to avoid while building applications.

By following OWASP Top 10, organizations can reduce security risks and protect user data better.